



Smart Crop Doctor

Scan Faster. Analyse Smarter. Prevents Loss

AI Intelligence for Precision Agriculture

Presented By: **Mercora International | Hyderabad, Telangana, India**

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HEADQUARTERS

A-212, Prime Legend, Botanical Garden Road,
Kondapur, Camelot Layout,
Hyderabad, Telangana 500084, India.

COMMUNICATIONS

info@mercorainternational.com
Contact No: +91 7680077222
www.mercorainternational.com



About Mercora International

A multi-sector company driving community-focused solutions for real-world challenges, with sustainability and inclusivity at its core..

MISSION

To create significant impact by empowering growth, innovation, and women entrepreneurship across.

VISION

To build a sustainable ecosystem through ethical practices, scalable innovation, and stakeholder collaboration”.

Key Team



Mrs. D. Jaya Geetha Kiran
*Founder &
Managing Partner*

Hails from agricultural family background, possess 15+ years cross-sector experience esp. in the field of Education, HR, Technology, Empowerment, etc. Former Partner at Sree Rakhadi Bio Pharmaceuticals. Expert in regulatory compliance & financial management. Master's in Computer Science. Participated in Viksit Bharat 2047 & ADB Conferences.



Ms. A. Manasa Raj
*Co Founder &
Partner*

Being Graduated from Delhi University, Passionate about women's entrepreneurship & Empowerment. A team builder and organizer of Awareness campaigns, cultural events. Works towards community engagement and development.

Our Journey to Smart Crop Doctor

First Insights - Awakening

Being from an agricultural family, seen sufferings of farmers and farm losses due to lack of information & timely detection of diseases. This has ignited and prompted me to come up with this concept of SMART CROP DOCTOR an based agritech Initiative as a ONE STOP SOLUTION to provide farmers timely and accurate information, advisories, prevent losses bringing a sense of complete hope, happiness and security.

Identifying the Problem

Through extensive outreach, identified significant crop losses from diseases, pests & insect infestations — which led to the development of Smart Crop Doctor, an AI-driven diagnostic tool for Precision Agriculture.

The Solution

Smart Crop Doctor leverages AI & drone-based technology to monitor soil and plant health, early detection of pest / insect attacks, diseases— enabling farmers to reduce losses & adopt sustainable agricultural practices.

Agriculture Challenges Today

**20–30%**

Crop losses due to pests & diseases

**40%+**

Excessive use of fertilizers & pesticides

**30%**

Water wastage in traditional irrigation



Growing labour shortages in rural farming



Lack of real-time farm insights & data

**Farmers rely on guesswork, not data.**

💡 AI + Drone-Based Smart Agriculture



Autonomous Drone Monitoring

Full-field scans replacing manual inspection in minutes



AI-Driven Crop Diagnostics

Real-time detection of disease, pests & deficiencies



Precision Spraying System

Targeted spray — up to 40% reduction in chemical use



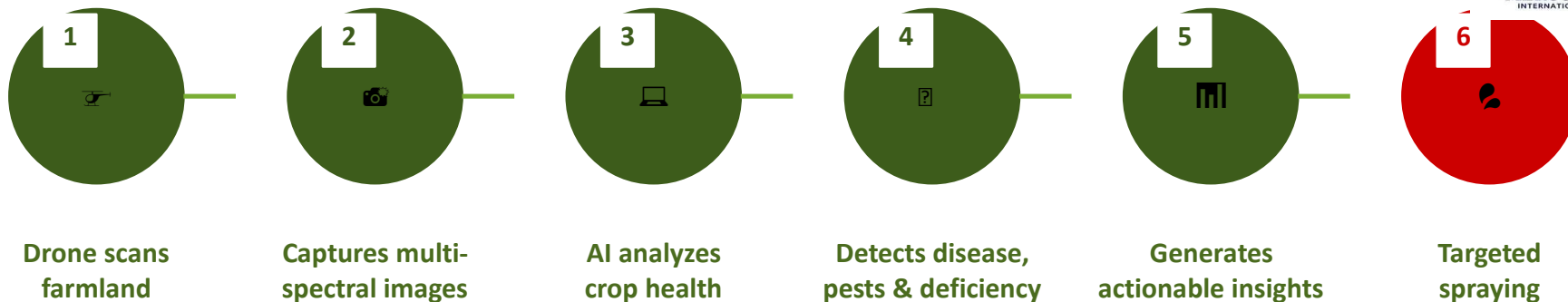
Real-Time Farm Analytics

Actionable insights delivered instantly to farmers

👉 **From guesswork to data-driven farming**



How It Works



AI Detection Capabilities:

 Soil nutrient deficiencies

 Early-stage crop diseases

 Pest-prone zones

 Thermal stress patterns



P10 Pro Drone Specs (Dastrionics)

 **Coverage:** 25 acres in 8 hrs (40 acres/day max)

 **Flight Time:** 3–4 acres per charge | Battery:
₹40,000

 **Altitude:** 508mm footprint — 40% more compact

 **GPS System:** RTK GPS — centimetre-level precision

 **Platform:** P10 Pro — 9.5 kg | 10L payload |
Made in India

 **Control:** Bike & shoulder portable — no pickup truck needed



Multispectral Camera — 5 Bands

RGB

450–680 nm

Visible color imagery — natural appearance

NIR

840 nm

Near-Infrared — NDVI chlorophyll mapping

Red
Edge

720 nm

Chlorophyll stress, early disease detection

Green

560 nm

GNDVI — dense canopy & biomass analysis

Red

650 nm

Plant stress, NDVI core band



15–25%

Increase in Crop Yield

Higher productivity per acre



30%

Reduction in Water Usage

Precision irrigation saves water



40%

Less Pesticide Use

Targeted spray, less chemical waste



↓35%

Lower Operational Costs

Automation replaces expensive labour



3×

Faster Farm Inspections

30 min vs 3 days for same coverage



Environment-Friendly Farming

Minimal soil disturbance and reduced carbon footprint through precision operations



Reduced Chemical Pollution

Targeted pesticide spraying cuts runoff into waterways and soil contamination



Efficient Resource Usage

Water, fertilizer and energy used only where needed — zero waste approach



Climate-Smart Agriculture

Supports climate-resilient cropping through data-driven adaptive farm management

140M+

Hectares of
Farmland
in India

<8%

Currently using
Precision
Agriculture

₹5,000Cr+

Agri-Drone Market
by 2027

Growth Drivers:

 Government Push for Digital Agriculture Mission & Drone Subsidies

 Rising smartphone & internet penetration among rural farmers

 FPO and cooperative-driven demand for precision agri services

 Integration with satellite data, IoT, and real-time weather feeds

 Huge untapped market — First-mover advantage: 2025–2030

Drone-as-a-Service (DaaS)



On-demand drone deployment for surveying & spraying — no asset ownership needed

Subscription Plans



Monthly/seasonal packages with unlimited surveys, analytics dashboard & expert support

Pay-Per-Acre Spraying



₹300–₹500 per acre per spray session — accessible for small & marginal farmers

Data Analytics Platform



B2B licensing to agri-input companies, banks & insurers for crop intelligence reports

Govt. & B2B Partnerships



State government contracts, FPO integrations, and CSR-funded deployments at scale



AI-Powered Disease Prediction

Predictive models flag crop stress 7–10 days before visible symptoms appear



Real-Time Decision Support

Instant alerts, recommendations & treatment plans pushed to farmer's phone



Scalable Drone Deployment

Fleet management system handles simultaneous multi-farm operations seamlessly



IoT & Satellite Data Integration

Combines ground sensors, drone imagery and satellite for full-spectrum intelligence

⚠️ Challenge

🔥 **High cost of drone equipment**

❓ 🌾 **Lack of awareness & digital literacy**

📋 **Regulatory compliance (DGCA norms)**

📶 **Connectivity issues in rural areas**

☁️ **Inconsistent weather / disruptions**

✅ Solution / Mitigation

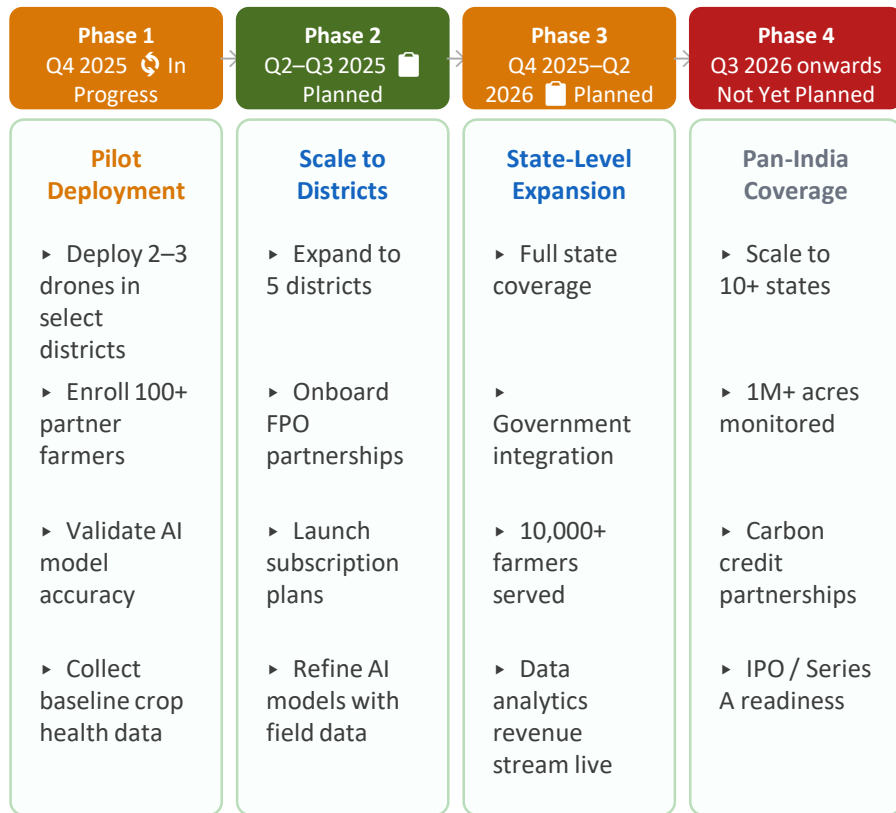
Shared drone-as-a-service model; fleet pooling across multiple farmers/FPOs

Farmer training programs, regional language app interface & local agri-agents

Built-in DGCA compliance workflows, geo-fencing, licensed pilot network

Offline AI inference models — analysis works without continuous internet

Automated rescheduling system and weather-aware flight planning algorithms



Seeking ₹X Crore Investment

35%

Drone Fleet Expansion

Procure 5+ additional drones for scaled deployment

25%

AI Model Development

Expand training datasets, improve prediction accuracy

20%

Market Penetration

Farmer outreach, agent network, regional partnerships

12%

Tech Infrastructure

Cloud, IoT sensors, data platform enhancements

8%

Compliance & Operations

DGCA licensing, insurance, admin setup



Documentary on Smart Crop Doctor



About This Documentary

A ground-level story of how Smart Crop Doctor is transforming Indian agriculture from farm visits to AI-driven insights.

Highlights

- ✓ Real farmer experiences & testimonials
- ✓ Drone operations in the field
- ✓ AI crop diagnostics live demo
- ✓ Impact on yield & farmer livelihoods
- ✓ Mercora's vision for pan-India scale

THANK YOU